



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°7 – Réseaux

COURS: Réseaux-2

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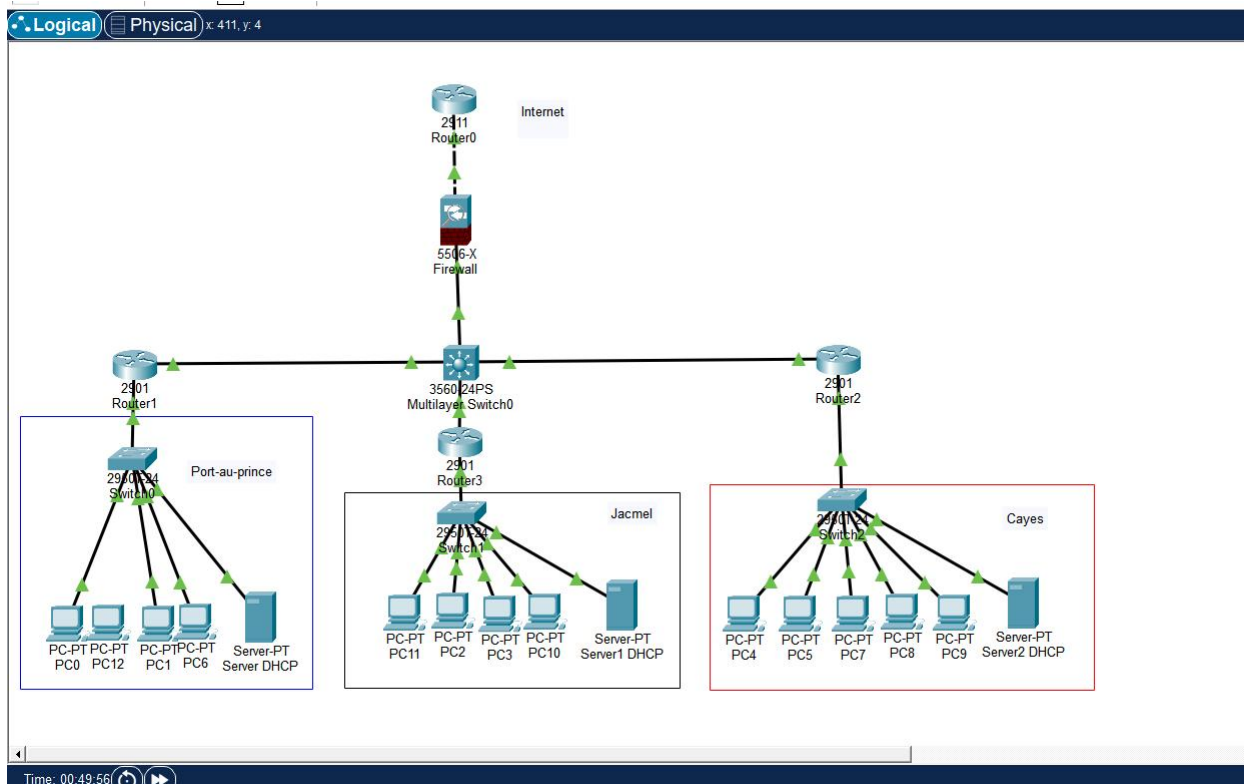
NIVEAU: L3

DATE: 05/06/2026

Objectifs

1. Configurer un réseau avec 3 routeurs
2. Mettre en place un Cisco ASA Firewall
3. Configurer le routage inter-sites (OSPF)
4. Créer une DMZ sécurisée
5. Implémenter des ACL sur ASA
6. Tester la sécurité du réseau
7. Configurer un VPN IPsec site-to-site
8. Sécuriser les communications entre 3 réseaux
9. Utiliser des routeurs Cisco (ou ASA si extension)

Sécurisation du réseau universitaire IUS avec Cisco ASA Firewall



Configuration de R1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Configuration de R2

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no sh

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-router)#

```

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Configuration de R3

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.10 255.255.255.252
Router(config-if)#no sh

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

```

Configuration de cisco ASA

```
.scoasa#conf t
.scoasa(config)#interface gigabitEthernet1/1
.scoasa(config-if)#nameif outside
.scoasa(config-if)#security-level 0
.scoasa(config-if)#ip address 203.0.113.2 255.255.255.0
.scoasa(config-if)#no shutdown
.scoasa(config-if)#interface gigabitEthernet1/2
.scoasa(config-if)#nameif inside
.scoasa(config-if)#security-level 100
.scoasa(config-if)#ip address 10.0.0.1 255.255.255.252
.scoasa(config-if)#no shutdown
.scoasa(config-if)#interface gigabitEthernet1/3
.scoasa(config-if)#nameif dmz
.scoasa(config-if)#security-level 50
.scoasa(config-if)#ip address 192.168.100.1 255.255.255.0
.scoasa(config-if)#no shutdown
```

```
ciscoasa(config-if)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
ciscoasa(config)#route inside 192.168.0.0 255.255.0.0 10.0.0.2
ciscoasa(config)#route dmz 192.168.100.0 255.255.255.0 192.168.100.10
```

NAT dynamique

```
ciscoasa(config)#object network INSIDE-NET
ciscoasa(config)#object network INSIDE-NET subnet 192.168.0.0 255.255.0.0
ciscoasa(config-network-object)#nat (inside,outside) dynamic interface
```

NAT statique

```
ciscoasa(config-network-object)#object network WEB-SERVER
ciscoasa(config-network-object)#host 192.168.100.10
ciscoasa(config-network-object)#nat (dmz,outside) static 203.0.113.10
```

ACL de sécurité

```
ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.20.0 255.255.255.0
192.168.100.0 255.255.255.0
ciscoasa(config)#access-list INSIDE-IN extended permit ip any any
ciscoasa(config)#access-group INSIDE-IN in interface inside
```

Routage internet sur ASA

```
ciscoasa(config)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
ciscoasa(config)#
```

Configuration du routeur Internet

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#interface g0/1
Router(config-if)#ip address 198.51.100.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

Router(config-if)#ip routing
Router(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.2
Router(config)#ip route 192.168.0.0 255.255.0.0 203.0.113.2
```

Conclusion

Grâce à ce TD j'ai pu savoir comment configurer un réseau avec 3 routeurs, créer une DMZ sécurisée et implémenter des ACL sur ASA.